

Volume 07 Development Guide - Development Guide Part 5

Version v40 draft

README.md

Development Guide Part 5

Version: v40 draft

Status: Draft

Document ID: MAL-V07-V40-README

Purpose

This release folder contains the official v40 documentation set for Development Guide Part 5.

Scope

- Covers DNA Development.
- Covers Documentation Workflow.
- Covers Version Management.

Acceptance

- All documents contain implementation-level guidance.
- The release PDF and ZIP are generated and verified.
- MANIFEST and RELEASE_NOTES are updated for v40.

Development Guide Part 5 Index

- [guide-07-40-01-dna-development.md](#) - DNA Development
- [guide-07-40-02-documentation-workflow.md](#) - Documentation Workflow
- [guide-07-40-03-version-management.md](#) - Version Management

DNA Development

Title: DNA Development

Version: v40 draft

Status: Draft

Document ID: MAL-V07-DNA-DEVELOPMENT-01

Purpose

Provide official implementation guidance for DNA Development inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V07-DNA-DEVELOPMENT-01	Governs DNA Development.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v40_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v40 draft	2026-06-29	Draft	Added official documentation for DNA Development.

Documentation Workflow

Title: Documentation Workflow

Version: v40 draft

Status: Draft

Document ID: MAL-V07-DOCUMENTATION-WORKFLOW-02

Purpose

Provide official implementation guidance for Documentation Workflow inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
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Acceptance Criteria

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- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
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- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V07-DOCUMENTATION-WORKFLOW-02	Governs Documentation Workflow.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v40_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v40 draft	2026-06-29	Draft	Added official documentation for Documentation Workflow.

Version Management

Title: Version Management

Version: v40 draft

Status: Draft

Document ID: MAL-V07-VERSION-MANAGEMENT-03

Purpose

Provide official implementation guidance for Version Management inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
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- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V07-VERSION-MANAGEMENT-03	Governs Version Management.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v40_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v40 draft	2026-06-29	Draft	Added official documentation for Version Management.